Fiscal Theory of the Price Level

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Consider the following fiscal-monetary model. It illustrates how the interaction between fiscal and monetary policy affects the price level. Reading the following paper may be helpful:

Leeper, E. M. (1991). Equilibria under 'active' and 'passive' monetary and fiscal policies. *Journal of Monetary Economics*, 27(1), 129-147.

Unless otherwise stated, variables without a subscript are assumed to be exogenous and constant, so you can treat them as parameters, rather than endogenous variables.

Households choose consumption C_t, real money balances M_t/P_t, real government bonds B_t/P_t to maximize:

$$\max_{C_t, M_t/P_t, B_t/P_t} E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\theta} - 1}{1-\theta} + \frac{(M_t/P_t)^{1-\mu} - 1}{1-\mu} \right)$$

subject to

$$C_t + \frac{M_t}{P_t} + \frac{B_t}{P_t} + T_t = I_{t-1}\frac{B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} + Y$$

where T_t denotes taxes (minus transfers) that the government raises from households. Y is exogenous income. I_t is the nominal interest rate.

The government budget constraint is given by:

$$\frac{M_t}{P_t} + \frac{B_t}{P_t} + T_t = I_{t-1}\frac{B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} + G$$

The government chooses government spending G and policies for T_t and M_t; where both T_t and M_t can follow some rules.

Assume that the government follows rules for taxes and for money supply of the following form

$$\frac{T_t}{Y} = \tau \frac{B_{t-1}/P_{t-1}}{Y} + v_t^\tau$$

$$M_t = M \exp(v_t^m)$$

where

$$v_t^\tau = \rho_\tau v_{t-1}^\tau + \varepsilon_t^\tau$$

$$v_t^m = \rho_m v_{t-1}^m + \varepsilon_t^m$$

and ε_t^τ and ε_t^m are fiscal and monetary shocks, zero mean, 1% standard deviation. Assume $\tau > (1/\beta - 1)$.

The parameters are calibrated at quarterly frequency. Use the following facts to calibrate some of the parameters in the model:

- Experimental evidence shows that the elasticity of intertemporal substitution equals $\frac{2}{3}$. Here the elasticity of intertemporal substitution is defined as a measure of responsiveness of the growth rate of consumption to the percentage change in the gross real interest rate.
- The average value of the nominal interest rate is 2.02% (at annual frequency).
- Experimental evidence shows that the elasticity of real money balances with respect to the percentage change in the net real interest rate is 0.24875.

Assume $\tau = 0.1$. Moreover, the persistence of all shocks equals 0.8 (i.e. $\rho_\tau = \rho_m = 0.8$). Also set $Y = 100$ and $G = 25$.

(a) First-order conditions and interpretation

First-order conditions

The first-order conditions from the household's optimization problem are:

Euler equation:

$$C_t^{-\theta} = \beta E_t \left[C_{t+1}^{-\theta} \frac{I_t}{P_{t+1}/P_t} \right]$$

Money demand equation:

$$\frac{(M_t/P_t)^{-\mu}}{C_t^{-\theta}} = 1 - \frac{1}{I_t}$$

Rearranging:

$$\frac{M_t}{P_t} = C_t^{\theta/\mu} \left(1 - \frac{1}{I_t} \right)^{-1/\mu}$$

Economic interpretation

How do real money balances depend on the nominal interest rate?

From the real money balances demand equation:

$$\frac{M_t}{P_t} = C_t^{\theta/\mu} \left(1 - \frac{1}{I_t} \right)^{-1/\mu}$$

we can see that if I_t rises $\Rightarrow (1-1/I_t)$ increases $\Rightarrow (1-1/I_t)^{-1/\mu}$ decreases $\Rightarrow \frac{M_t}{P_t}$ decreases.

So, there is a **negative dependence**: higher nominal interest rate $\Rightarrow$ lower real money balances.

Effect of expected inflation on the nominal interest rate:

From the Euler equation:

$$C_t^{-\theta} = \beta E_t \left[C_{t+1}^{-\theta} \frac{I_t}{P_{t+1}/P_t} \right]$$

Define inflation $\pi_{t+1} = P_{t+1}/P_t$. Then:

$$\frac{I_t}{E_t[\pi_{t+1}]} = \frac{C_t^{-\theta}}{\beta E_t[C_{t+1}^{-\theta}]}$$

Holding consumption constant, the real interest rate is determined by the discount factor. Therefore, a rise in expected inflation leads to a one-for-one increase in the nominal interest rate. This is the Fisher effect: expected inflation has a **positive effect** on the nominal interest rate.

(b) Equilibrium

A set of prices $\{P_t\}$ and allocations $\{C_t, M_t/P_t, B_t/P_t, I_t, T_t\}$ satisfying:

(1) Euler equation: $C_t^{-\theta} = \beta E_t \left[C_{t+1}^{-\theta} \frac{I_t}{P_{t+1}/P_t} \right]$

(2) Money demand: $\frac{M_t}{P_t} = C_t^{\theta/\mu} \left(1 - \frac{1}{I_t}\right)^{-1/\mu}$

(3) Household budget constraint: $C_t + \frac{M_t}{P_t} + \frac{B_t}{P_t} + T_t = I_{t-1}\frac{B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} + Y$

(4) Government budget constraint: $\frac{M_t}{P_t} + \frac{B_t}{P_t} + T_t = I_{t-1}\frac{B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} + G$

(5) Tax rule: $\frac{T_t}{Y} = \tau \frac{B_{t-1}/P_{t-1}}{Y} + v_t^\tau$

(6) Money supply rule: $M_t = M \exp(v_t^m)$

(7) Market clearing: $C_t + G = Y$

(8) Shock processes: $v_t^\tau = \rho_\tau v_{t-1}^\tau + \varepsilon_t^\tau$, $v_t^m = \rho_m v_{t-1}^m + \varepsilon_t^m$

The endogenous variables are C_t, M_t/P_t, B_t/P_t, I_t, T_t, P_t, M_t. The shocks v_t^τ and v_t^m follow AR(1) processes with persistence $\rho = 0.8$.

(c) Steady state derivation

Calibrated parameters

Using the calibrated parameter values:

$$\begin{aligned} \theta &= 1.5 \\ \mu &= 4.0 \\ \beta &= 0.995 \\ \tau &= 0.1 \\ Y &= 100 \\ G &= 25 \\ M &= 10 \end{aligned}$$

Steady state equations

In steady state, all variables are constant. Let variables without time subscript denote steady state values.

(1) From market clearing:

$$C = Y - G = 100 - 25 = 75$$

(2) From Euler equation:

$$I = \frac{1}{\beta} = \frac{1}{0.995} = 1.005025$$

(3) From money demand:

$$m = \frac{M}{P} = C^{\theta/\mu}\left(1 - \frac{1}{I}\right)^{-1/\mu}$$

(4) From tax rule:

$$T = \tau \frac{B}{P}$$

(5) From government budget constraint:

$$\frac{M}{P} + \frac{B}{P} + T = I\frac{B}{P} + \frac{M}{P} + G$$

Simplifying:

$$\frac{B}{P} + T = I\frac{B}{P} + G$$

$$T = (I-1)\frac{B}{P} + G$$

Combining with tax rule:

$$\tau\frac{B}{P} = (I-1)\frac{B}{P} + G$$

$$(\tau - (I-1))\frac{B}{P} = G$$

$$\frac{B}{P} = \frac{G}{\tau - (I-1)}$$

Since $\tau > 1/\beta - 1 = I - 1$ by assumption, the denominator is positive.
(6) Then taxes:

$$T = \tau\frac{B}{P}$$

(7) Price level from money supply:

$$P = \frac{M}{m}$$

Numerical steady state results

The steady state results for this economy in levels:

$$
\begin{aligned}
C^{SS} &= 75.0000 \\
m^{SS} &= 18.9448 \\
b^{SS} &= 263.3462 \\
I^{SS} &= 1.0051 \\
T^{SS} &= 26.3346 \\
M^{SS} &= 10.0000 \\
P^{SS} &= 0.0528 \\
v_{\tau}^{SS} &= 0 \\
v_{m}^{SS} &= 0
\end{aligned}
$$

Steady state results for this economy in logs:

$$
\begin{aligned}
\ln C^{SS} &= 4.31749 \\
\ln m^{SS} &= 2.94153 \\
\ln b^{SS} &= 5.57347 \\
I^{SS} &= 0.005055 \\
\ln T^{SS} &= 3.27088 \\
\ln M^{SS} &= 2.30259 \\
\ln P^{SS} &= -0.63894
\end{aligned}
$$

Stability analysis

The eigenvalue analysis shows two roots larger than unity in modulus, which matches the number of forward-looking variables (C and P), and the rank condition is verified. This ensures the existence of a unique stable equilibrium (Blanchard-Kahn condition is satisfied).

(d) Impulse responses

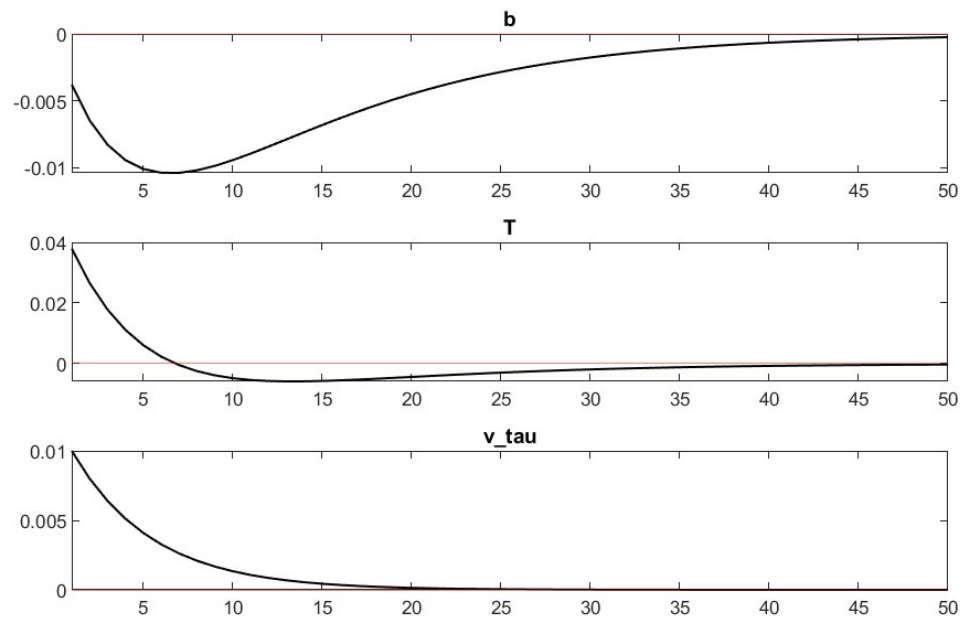

Figure 1: Impulse responses to a 1 percent positive shock to v^τ over 50 periods

Response to fiscal shock (v^τ)

In response to a 1% positive fiscal shock, the fiscal disturbance v^τ immediately shows 1% positive deviation relative to its steady state. The disturbance then gradually decays back to steady state.

As the fiscal shock directly leads to a 1% positive deviation of v^τ from the steady state, it raises taxes by $Y \times 0.01 = 100 \times 0.01 = 1$ unit in levels. Relative to the steady-state tax level of about 26.33, this corresponds to a 3.8% positive deviation from the steady state. This immediate increase in tax improves the government's fiscal position. Through the government budget constraint, higher taxes on the left-hand side reduce the need for new borrowing, so real debt begins to fall right from the moment when the shock occurs: -0.5% deviation from the steady state.

As the fiscal disturbance decays gradually (with persistence $\rho = 0.8$), taxes remain above their steady-state level for several periods. This sustained situation continues to put downward pressure on debt.

Once the direct effect of the shock fades, taxes become increasingly determined by the second term in the tax rule which depends on real debt. As real debt is below its steady-state level, this term pulls taxes below their steady state. Taxes show a slight negative deviation from the steady state after about period 7. The government budget constraint now works in reverse, with taxes below steady state, the government must increase borrowing to finance its constant spending. As a result, real debt starts returning to its steady state.

As for the price level and interest rates, the fiscal shock does not enter the equations for real money balances demand or the Euler equation to affect them. The money supply

follows a process determined by the constant M and the monetary shock, so it is unaffected by the fiscal shock. Consequently, the price level and the nominal interest rate remain at their steady-state levels.

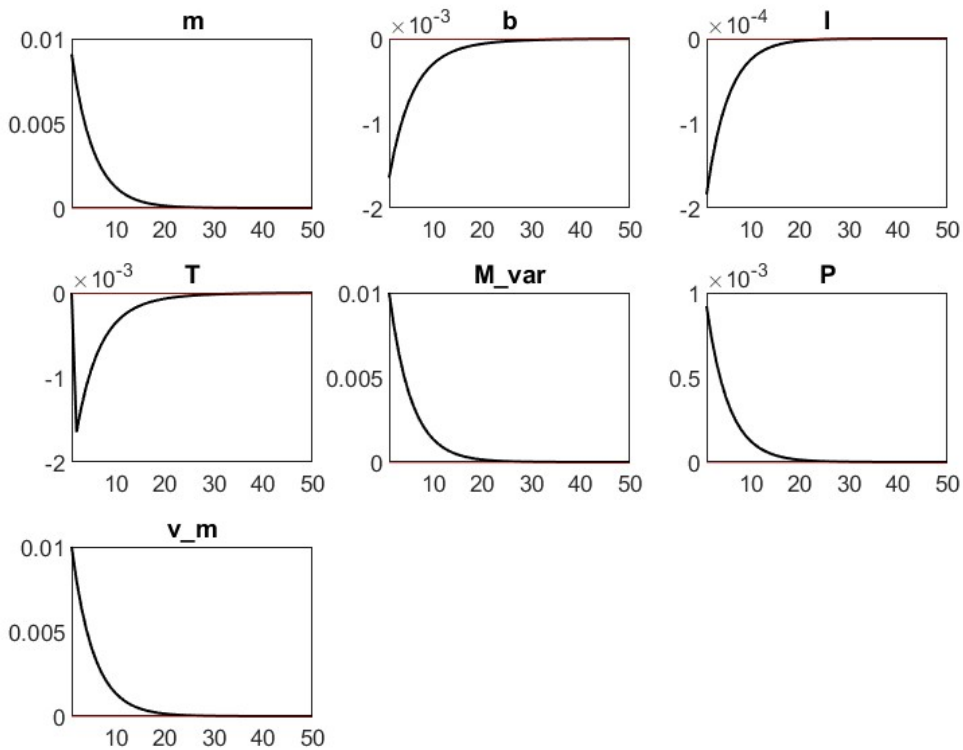

Figure 2: Impulse responses to a 1 percent positive shock to v^m over 50 periods

Response to monetary shock (v^m)

In response to a 1% positive monetary shock, the monetary disturbance v^m immediately shows 1% positive deviation relative to its steady state. The disturbance then gradually decays back to steady state.

The shock has a direct effect on the monetary supply. The positive 1% monetary shock results in the immediate 1% positive deviation from the steady state and then it returns slowly to the steady state.

As for the price level, a positive 1% monetary shock leads to the immediate nominal money supply positive deviation from the steady state. In response, the price level jumps instantly by about 0.1% above its steady state. This adjustment reflects the equilibrium in the money market: with higher nominal money supply, the price level rises to clear the market. After the initial positive deviation from the steady state, the price level gradually returns to its steady state as the shock is temporary.

In response to a 1% positive monetary shock, real money balances show an initial positive deviation of about 0.9% from the steady state. From the money demand equation, consumption is constant, so a positive deviation of real money balances from a steady state requires a decrease in the nominal interest rate. With $I - 1 = 0.005$ and parameter $\mu = 4$, a 1%

positive deviation from the steady state for m implies a -0.002 percentage points change in I. The IRF shows I falling by 0.02 percentage points.

In response to a 1% positive monetary shock, the price level reacts instantly with a positive percentage deviation from the steady state, leading to a deviation in real debt. As a result, real debt shows a roughly -0.18% deviation in period 0. This decline in debt then affects taxes with a one-period lag through the tax rule. In the period following the shock, taxes exhibit a negative deviation of about -0.15%.

Policy classification

According to the Leeper classification, monetary policy is **active** because the money supply follows a rule that does not respond to the level of government debt. Fiscal policy is **passive**, as taxes follow a rule where they respond directly to the lagged real debt level. The condition $\tau > 1/\beta - 1$ ensures that this response is strong enough to stabilize debt over time.

According to Leeper, the combination of active monetary policy and passive fiscal policy ensures a unique stable solution for the model. In this regime, fiscal shocks do not affect the price level or interest rates, and government debt is stabilized by tax adjustments.

(e) Passive Monetary Policy / Active Fiscal Policy

Consider the case now where taxes are exogenous

$$\frac{T_t}{Y} = \tau \frac{B_{ss}/P_{ss}}{Y} + v_t^\tau$$

where B_{ss} is the steady-state value of bonds, and $P_{ss} = 0.2$ is the new steady-state price level. The money supply M_t rule above is replaced by the interest rate rule of the form:

$$I_t = \frac{1}{\beta}\left(\frac{P_t}{P_{ss}}\right)^\alpha / \exp(v_t^m)$$

where $\alpha = -1/2$. The value of M should be now chosen endogenously to ensure that $P = P_{ss}$ in the steady state.

Steady State in the New Framework

In the steady state, all real variables are unaltered compared to the results in point (c). The difference is that in point (c) M was exogenously given, but in point (e) P is exogenously given. In point (c) P adjusted to ensure that $M = M^{ss}$, but in point (e) M adjusts to ensure that $P = P_{ss}$.

Steady state results:

$$C = Y - G = 75$$

$$I = \frac{1}{\beta} = 1.005025$$

$$\frac{B}{P} = \frac{G}{\tau - (I-1)} = 263.3462$$

$$T = \tau \frac{B}{P} = 26.3346$$

$$m = C^{\theta/\mu} \left(1 - \frac{1}{I}\right)^{-1/\mu} = 18.9448$$

$$P = 0.2 \quad \text{(given)}$$

$$M = m \cdot P = 18.9448 \times 0.2 = 3.78896$$

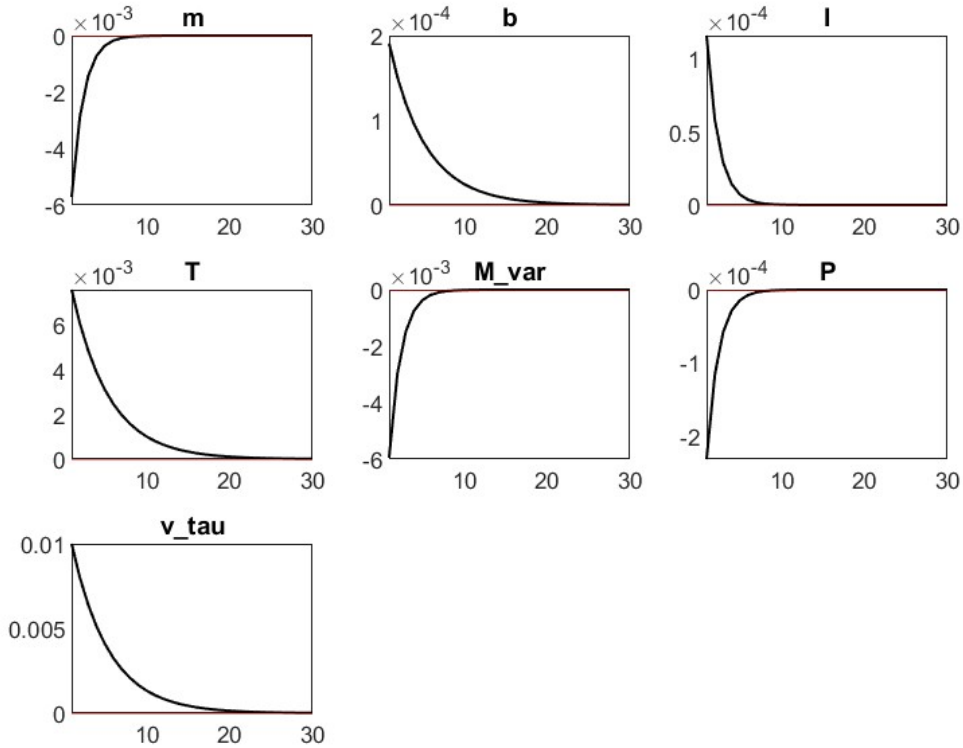

Figure 3: Impulse responses to a 1 percent positive shock to v^τ over 30 periods

Response to Fiscal Shock (v^τ)

A positive 1% shock to ε^τ causes a positive deviation from the steady state for fiscal disturbance v^τ by 1% in period $t = 0$. The disturbance then decays gradually with persistence $\rho = 0.8$.

Taxes: react immediately at $t = 0$ to the 1% positive fiscal shock. The tax rule links taxes to two components: a fixed part determined by the steady-state levels of debt and prices, and a variable part proportional to the fiscal disturbance. The 1% shock raises the level of taxes by 1 unit. Relative to steady-state taxes (26.32), this is about 3.8% deviation from the steady state.

Price level: shows a negative roughly 1.7% deviation from the steady state at $t = 0$ and then gradually returns to steady state by period 10. Change in the price level is a key adjustment mechanism. When taxes rise, the government budget constraint must still hold. With taxes fixed, the only way to satisfy this constraint is through changes in the price level. A lower price level increases the real value of past government debt and money, which helps balance the government budget constraint.

Interest rate: shows a positive about 0.9 percentage points deviation from the steady state in response to the shock at $t = 0$ and then returns to steady state by period 10. This happens because of the interest rate rule that reacts to the price level with a negative coefficient ($\alpha = -0.5$). When the price level falls, the interest rate rule leads to a higher interest rate.

Real money balances: have a strong negative deviation from the steady state by about

45% at period $t = 0$ and then recover. This large decrease is explained by two factors. First, the nominal interest rate value is very low in the steady state, so even a small increase makes bonds much more attractive relative to money. Second, the demand for money is very sensitive to changes in the interest rate due to the calibration.

Nominal money supply: also shows a negative percentage deviation from the steady state by about 48% in period $t = 0$ as a response to the positive fiscal shock and then returns to the steady state. This happens as the money supply now adjusts endogenously.

Real government debt: has a positive deviation by about 5% from the steady state in response to the positive fiscal shock and then gradually returns to the steady state. When the price level falls, the real value of past nominal debt increases, raising the right-hand side of the government budget constraint.

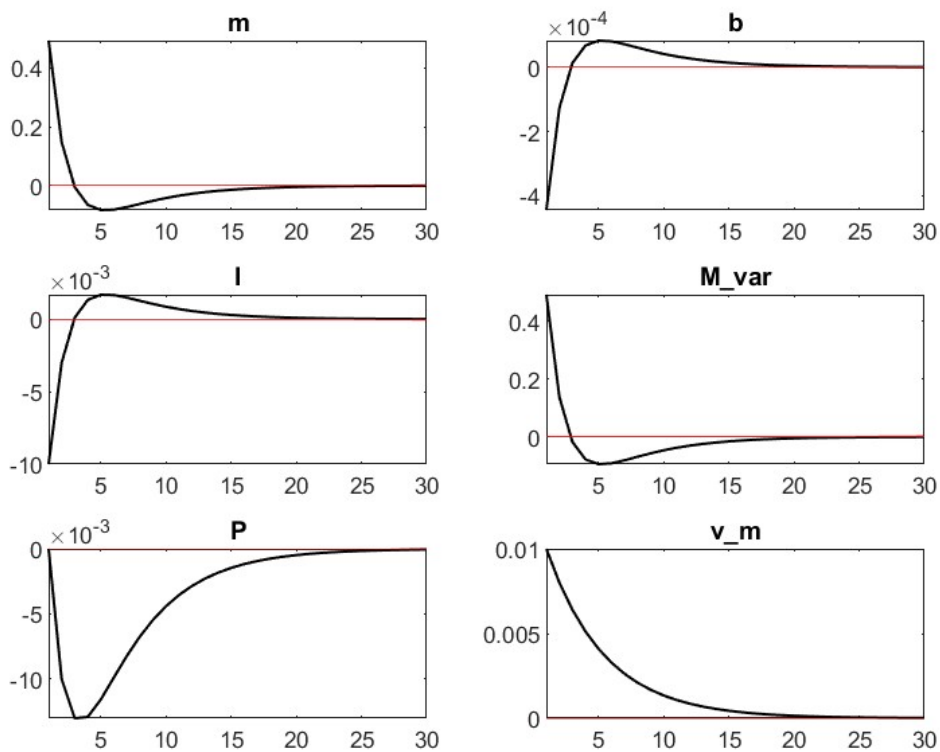

Figure 4: Impulse responses to a 1 percent positive shock to v^m over 30 periods

Response to Monetary Shock (v^m)

In response to a 1% positive monetary shock, the monetary disturbance v^m immediately shows 1% positive deviation relative to its steady state. The disturbance then gradually decays back to steady state.

Nominal interest rate: has a negative immediate 1 percentage point deviation from the steady state. This happens because the shock is built directly into the interest rate rule. When the monetary disturbance goes up, the rule pushes the interest rate down.

For the next few periods, the interest rate stays below the steady state. Then around period 4, prices have dropped about 1.2% below the steady state. As the interest rate rule

reacts negatively to prices ($\alpha = -0.5$), this drop in prices pushes the interest rate up. After that, both prices and the interest rate slowly go back to steady state.

Price level: does not react to the shock in the first period. It starts having a negative percentage deviation from the steady state in the next period, reaches a minimum of about 1.2% below steady state around period four, and then slowly recovers.

Real money balances: show about 50% positive deviation from the steady state as a response to the monetary shock. When the interest rate falls, the term that determines money demand increases sharply. After the initial jump, real money balances start declining. Around period four, they become slightly negative, then slowly return to steady state.

Nominal money supply: follows a path very similar to real money balances.

Real government debt: has an about -4% deviation from steady state as a response to the monetary shock. Around period four, real debt becomes slightly positive, then slowly returns to steady state.

Taxes: do not react to the monetary shock because they do not depend on any variables affected by the shock.

Comparison between Part (d) and Part (e)

The impulse responses to a fiscal shock in part (e) look very different from those in part (d).

In part (d), fiscal policy is passive (taxes react to past debt). Monetary policy is active (the money supply is fixed and doesn't respond to debt). When a fiscal shock occurs: taxes go up, debt goes down, but prices, interest rates, and real money balances don't move at all. Debt comes back to normal simply because taxes will adjust in the future.

In part (e), it's the opposite. Fiscal policy is active (taxes are fixed and don't depend on debt). Monetary policy is passive (the interest rate rule responds to prices). When a fiscal shock occurs: taxes go up, prices fall, interest rates rise, real money balances fall sharply, and debt actually increases. Here, it's not taxes that bring debt back, it's the price level doing so.

Policy Classification for Part (e)

According to the Leeper classification:

- **Fiscal policy: active** — taxes do not respond to the level of government debt.
- **Monetary policy: passive** — its reaction to the price level indirectly helps stabilize the real value of government debt through the government budget constraint.

(f) Sensitivity Analysis for α Parameter

If $\alpha = \frac{1}{2}$, it is impossible to solve this model as Blanchard & Kahn conditions are not satisfied: no stable equilibrium as there are 2 jump variables, but 3 unstable eigenvalues.

Let's first examine the reaction of the model to a fiscal shock. The key transmission mechanism works as follows: when taxes increase due to a fiscal shock, the government budget constraint must be satisfied. This creates pressure for the price level to adjust, which then feeds back into the interest rate through the policy rule, affecting money demand and

then the government budget. The stability of the system depends crucially on the strength and direction of this feedback loop.

In the case of $\alpha = \frac{1}{2}$, and as my other trials show all other $\alpha > 0$, following a fiscal shock, taxes rise and prices fall to satisfy the government budget constraint. Because α is positive, the fall in prices causes the interest rate to decrease. A lower interest rate increases the demand for real money balances, as holding money becomes cheaper. With higher money demand, the price level must fall further to maintain equilibrium in the money market. At the same time, the government budget constraint is also affected: the initial fall in prices has already increased the real value of past debt, and further price changes will continue to influence the required adjustment in real debt. However, the key destabilizing mechanism comes from the money market: falling prices reduce interest rates (when $\alpha > 0$), which increases money demand, requiring even lower prices. There is no opposing force to reverse the decline, and the system drives away from steady state. So, the interest rate rule directly targets equilibrium in the money market by adjusting the nominal rate in response to price changes. However, because real money balances enter the government budget constraint, monetary policy indirectly affects fiscal solvency. For a stable equilibrium, the rule must balance the money market in a way that also helps satisfy the government budget constraint.

In previous part (e) $\alpha = -0.5$ which ensures a unique solution to the model. It is so because in response to the fiscal shock monetary policy provides a stabilizing force. When a positive fiscal shock raises taxes, the government budget constraint requires an adjustment. The price level falls, which increases the real value of past nominal debt and helps balance the budget. Because α is negative but moderate, this fall in prices leads to a moderate increase in the interest rate through the policy rule. The higher interest rate reduces the demand for real money balances, which in turn puts upward pressure on prices to restore money market equilibrium. This creates a negative feedback loop: falling prices raise rates, which reduces money demand, which pushes prices back up. The system returns to steady state smoothly.

After several trials, I have found out that when roughly $-2 < \alpha < 0$, there is a unique solution to the model. When $\alpha < -2$, Blanchard-Kahn conditions are not satisfied, there is no solution to the model, as extra unstable eigenvalue appears. But when $-2 < \alpha < -1$, the system becomes oscillatory but still stable. A fiscal shock causes prices to fall, and with α strongly negative, this generates a sharp increase in the interest rate. The rate growth reduces money demand. To restore money market equilibrium, prices must now rise. Rising prices, through the strongly negative rule, cause the interest rate to fall sharply again. This creates a cycle of overshooting: prices fall, rates spike, prices rise, rates decrease, and so on. The economy oscillates around steady state, and there is a solution to the model, but IRFs look like shown on Figure 5. If α is too negative (for example -2), the oscillations don't fade away, they actually get bigger over time, so the economy never stabilizes. The system develops an additional unstable eigenvalue, and no stable equilibrium exists.

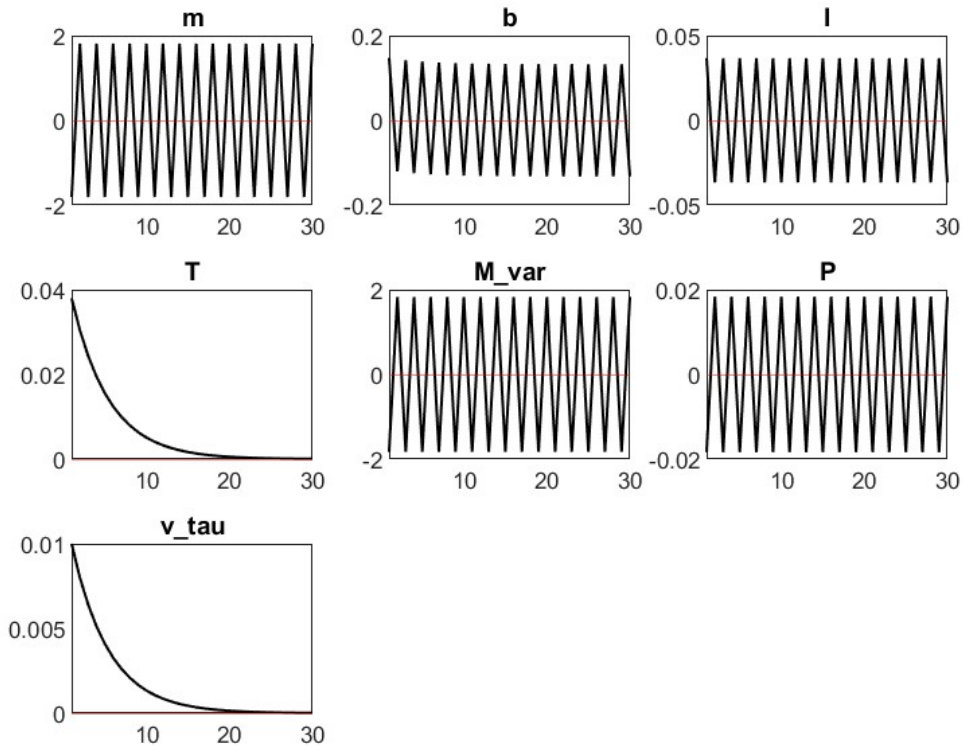

Figure 5: IRFs for all variables to a 1 percent positive shock to v^{τ} over 30 periods when $-2 < \alpha < -1$

When $\alpha = 0$, the interest rate becomes completely unresponsive to the price level. After a fiscal shock, taxes rise and the price level falls, but it has no effect on the interest rate. The interest rate remains at its steady-state level, determined only by the discount factor and the monetary shock. Consequently, there is no feedback from prices to money demand through the interest rate. The price level is now determined solely by the government budget constraint, but this is one equation with two unknowns (prices and debt). The system becomes underdetermined, the Jacobian becomes singular, and no unique steady state exists.

So, we can say that there is a unique and stable solution to the model when $-2 < \alpha < 0$. And the dynamics of the variables in response to 1% positive fiscal shock is smooth without oscillations when $-1 < \alpha < 0$. Oscillatory behavior of IRFs for variables with the existence of the model's solution occurs when $-2 < \alpha < -1$. No solution exists when $\alpha = 0$, $\alpha \leq -2$, $\alpha \geq 0$.

For a monetary shock, the same logic applies, but the transmission starts with an immediate drop in the interest rate, which then triggers adjustments in money demand, prices, and the government budget constraint through the same feedback mechanisms determined by α.

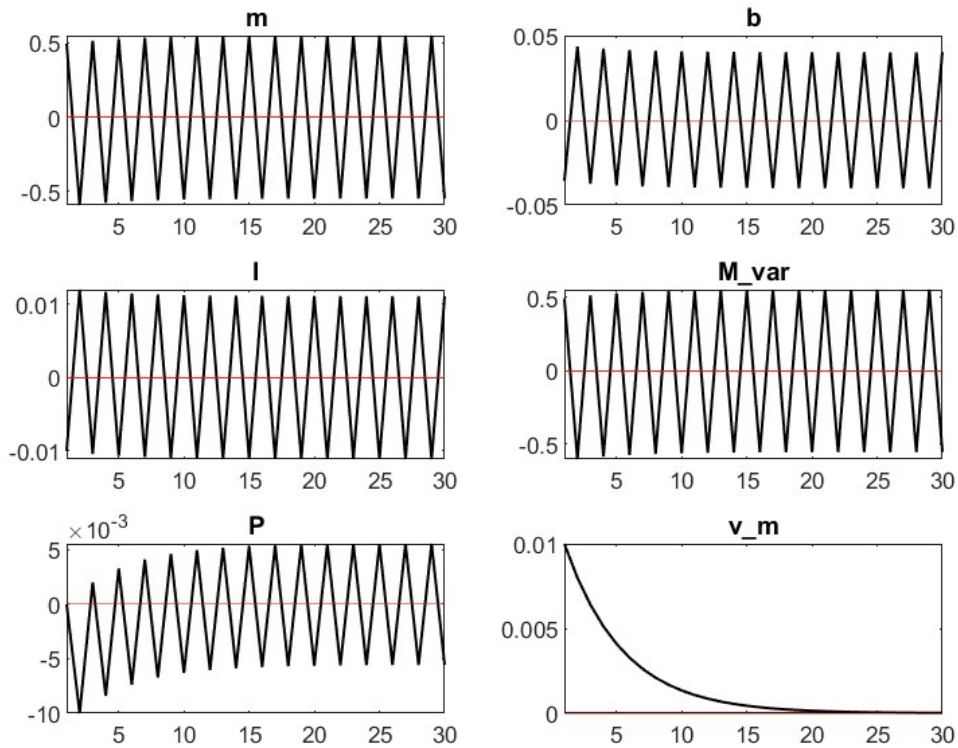

Figure 6: IRFs for all variables to a 1 percent positive shock to v^m over 30 periods when $-2 < \alpha < -1$

Summary of Stability Conditions

α **Range**	**Result**
$-1 < \alpha < 0$	Unique stable solution, smooth dynamics
$-2 < \alpha < -1$	Unique stable solution, oscillatory dynamics
$\alpha = 0$	No solution (singular Jacobian, indeterminacy)
$\alpha > 0$	No solution (system diverges)
$\alpha \leq -2$	No solution (explosive oscillations)

Table 1: Summary of existence and stability for different α values